# TRAVEL BEHAVIOUR ANALYTICS AND SMART TRIP RECOMMENDATION SYSTEM

A Project Report
Submitted in the partial fulfillment of the
requirements for the award of the degree of

## BACHELOR OF TECHNOLOGY

**In**

## DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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# Declaration

The Project Report entitled **"Travel Behaviour Analytics and Smart Trip Recommendation System"** is a record of Bonafide work of team members **Vamka Bhavya – 2320030319, Koneru Nikhil – 2320090025, and Beeraka Joshitha – 2320090048**, submitted in partial fulfilment for the award of B. Tech in Computer Engineering to K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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# Certificate

This is to certify that the project-based report entitled **"Travel Behaviour Analytics and Smart Trip Recommendation System"** is a Bonafide work carried out and submitted by **Vamka Bhavya – 2320030319, Koneru Nikhil – 2320090025, and Beeraka Joshitha – 2320090048**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science Engineering**, at **K L (Deemed to be University)**, during the academic year **2025–2026**.

**Signature of the Supervisor**

**Signature of the HOD** **Signature of the External Examiner**

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# TABLE OF CONTENTS


| S.No | Contents | Page no |
|---|---|---|
| 1 | Abstract | 6 |
| 2 | Introduction | 7 |
| 3 | Literature survey | 9 |
| 4 | Hardware and Software requirements | 11 |
| 5 | Implementation | 13 |
| 6 | Experimentation and Code | 16 |
| 7 | Results | 18 |
| 8 | Conclusion | 22 |
| 9 | References | 23 |

# ABSTRACT

In today's digital era, large volumes of travel data are generated through online booking systems and tourism platforms. Analyzing this data is essential for understanding travel patterns, cost distribution, and customer behavior. This project, titled "Travel Data Analysis and Recommendation System using Azure Machine Learning and Power BI," focuses on extracting meaningful insights from travel data using cloud-based technologies and visualization tools.

The project utilizes Microsoft Azure Machine Learning for data handling and model training, and Power BI for data visualization. The dataset includes attributes such as destination, travel duration, accommodation type, accommodation cost, transportation cost, and traveler details. During preprocessing, data cleaning techniques were applied to remove unnecessary symbols, handle missing values, and convert data into appropriate formats.

Automated Machine Learning (AutoML) was used to train models, and the best-performing model was identified as a Voting Ensemble model. Power BI was used to create interactive visualizations such as bar charts, pie charts, and maps to represent travel trends and cost analysis effectively.

The results highlight key insights such as peak travel months, most expensive destinations, and distribution of accommodation types. The project demonstrates that cloud-based analytics combined with visualization tools can efficiently process and analyze large datasets, supporting better decision-making.

Overall, this system provides a scalable and efficient approach to travel data analysis and can be further extended for real-time applications and recommendation systems.

# INTRODUCTION

In today's digital era, the travel and tourism industry generates a large amount of data through online booking platforms, travel agencies, and digital payment systems. This data includes information about destinations, travel duration, accommodation types, transportation costs, and traveler details. Analyzing such data helps in understanding travel trends, customer preferences, and cost patterns, which can support better decision-making.

Traditional data analysis methods are often insufficient to handle large-scale datasets efficiently. With the advancement of cloud computing and data analytics tools, it has become easier to process and analyze large volumes of data in a scalable and efficient manner. Cloud platforms provide high computational power, storage, and flexibility, making them suitable for modern data-driven applications.

This project, titled "Travel Data Analysis and Recommendation System using Azure Machine Learning and Power BI," focuses on analyzing travel data using cloud-based technologies. Microsoft Azure Machine Learning is used for managing datasets and training models, while Power BI is used for creating interactive dashboards and visualizations.

The project aims to extract meaningful insights such as travel trends, cost distribution, and duration patterns. By combining machine learning and visualization tools, the system provides a comprehensive approach to understanding travel data and presenting it in an easy-to-understand format.

## Problem Description:

The travel industry generates large volumes of data from various sources such as online bookings, travel agencies, and digital transactions. Managing and analyzing this data using traditional methods is challenging, time-consuming, and inefficient. As the size of the dataset increases, it becomes difficult to identify meaningful patterns, trends, and relationships within the data.

One of the major challenges is understanding cost variations across destinations, accommodation types, and transportation methods. Additionally, identifying peak travel periods and analyzing customer preferences requires proper data processing and visualization techniques. Without effective tools, it is difficult to extract actionable insights from raw data.

Another problem is the lack of integration between data analysis and visualization systems. Raw data alone does not provide meaningful insights unless it is processed and represented visually. Therefore, there is a need for a system that can efficiently handle large datasets, perform analysis, and present the results in a clear and interactive manner.

To address these challenges, this project uses cloud-based technologies and visualization tools to analyze travel data efficiently and generate useful insights.

## Objectives:

The main objectives of this project are:

• To analyze travel data and extract meaningful insights
• To identify travel trends such as peak travel months and popular destinations
• To study cost distribution based on accommodation and transportation
• To analyze travel duration patterns across different categories
• To visualize data using Power BI dashboards for better understanding
• To use Azure Machine Learning for dataset management and model training
• To develop an efficient and scalable system for travel data analysis

The project aims to provide a clear understanding of travel patterns and support better decision-making using data-driven insights

# LITERATURE SURVEY

The travel and tourism industry has increasingly adopted data analytics and machine learning techniques to understand customer behavior, optimize costs, and improve decision-making processes. Various studies have highlighted the importance of analyzing travel data to identify patterns related to destinations, expenses, and user preferences.

According to recent research, data analytics plays a crucial role in transforming raw travel data into meaningful insights. By applying statistical methods and machine learning algorithms, organizations can predict travel trends, recommend destinations, and optimize pricing strategies. Techniques such as classification, clustering, and regression are commonly used to analyze travel-related datasets.

Cloud-based platforms like Microsoft Azure Machine Learning have gained popularity due to their scalability, flexibility, and ability to handle large datasets efficiently. These platforms provide automated machine learning (AutoML) capabilities, which allow users to build and evaluate models without extensive programming knowledge. This makes them suitable for students and professionals working on data-driven projects.

Visualization tools such as Microsoft Power BI are widely used to represent data in an interactive and user-friendly manner. Studies show that visual dashboards help in better understanding of complex datasets by converting numerical information into charts, graphs, and maps. Features like filtering, drill-down analysis, and real-time updates enhance decision-making capabilities.

Several research works have also focused on travel recommendation systems, where machine learning models are used to suggest destinations based on user preferences, historical data, and cost analysis. These systems improve user experience by providing personalized recommendations and insights.

Despite the advancements, challenges such as data preprocessing, handling missing values, and integrating different tools still exist. However, combining cloud computing, machine learning, and visualization tools provides an effective solution for travel data analysis.

This project builds upon these existing studies by integrating Azure Machine Learning for data processing and model training with Power BI for visualization, thereby creating a comprehensive system for analyzing travel data and generating meaningful insights.

# HARDWARE AND SOFTWARE REQUIREMENTS

## Hardware Requirements

The Travel Data Analysis and Recommendation System is a cloud-based project, and most of the processing is performed using Microsoft Azure services. Therefore, high-end local hardware is not required. However, a basic system configuration is needed to access cloud platforms and visualization tools smoothly.

**Processor:**

A system with an Intel Core i3 processor or equivalent is sufficient. For better performance, an Intel Core i5 or higher is recommended.

**RAM:**

A minimum of 4 GB RAM is required to run Power BI and web browsers efficiently. For smoother performance, 8 GB RAM is recommended.

**Storage:**

At least 500 MB of free disk space is required for storing project files, temporary data, and application dependencies.

**Graphics:**

No dedicated graphics card is required. Integrated graphics are sufficient for data visualization tasks.

**Operating System:**

The project can be executed on any operating system such as Windows, Linux, or macOS, as it mainly relies on cloud-based tools.

**Software Requirements**

The project uses cloud-based tools and data visualization software to perform analysis efficiently.

**Cloud Platform:**

• Microsoft Azure Machine Learning – Used for dataset management and model training

• Azure Blob Storage – Used for storing dataset

**Visualization Tool:**

• Microsoft Power BI – Used for creating interactive dashboards and visualizations

**Programming / Data Format:**

• CSV Dataset – Used as input data format

**Development Tools:**

• Web Browser (Google Chrome / Microsoft Edge) – Used to access Azure and Power BI

• Power Query Editor – Used for data cleaning and transformation

With the above hardware and software configuration, the system can be executed efficiently. Since most of the processing is performed in the cloud, the project ensures scalability, flexibility, and minimal dependency on local resources.

# IMPLEMENTATION

Implementation of Travel Data Analysis System

The implementation of the Travel Data Analysis and Recommendation System focuses on analyzing travel-related data using cloud-based technologies and visualization tools. The system is designed to store, process, analyze, and visualize travel data efficiently using Microsoft Azure Machine Learning and Power BI.

The implementation process is divided into several stages, including data collection, data preprocessing, model training, and data visualization.

**1. Dataset Description and Data Storage**

The dataset used in this project contains travel-related information such as destination, start date, end date, travel duration, accommodation type, accommodation cost, transportation type, transportation cost, and traveler details. The dataset is stored in CSV format.

Initially, the dataset is uploaded to Azure Blob Storage, which provides a secure and scalable environment for storing large datasets. Azure Blob Storage allows easy access and integration with Azure Machine Learning.

**2. Data Asset Creation in Azure ML**

After uploading the dataset, a data asset is created in Azure Machine Learning Studio. The dataset is converted into a tabular format, enabling structured analysis and model training. Proper configuration of the dataset ensures that each column is recognized correctly for further processing.

### 3. Data Preprocessing

Before analysis, the dataset undergoes preprocessing to improve data quality. The preprocessing steps include:

• Removing special characters such as currency symbols from cost columns
• Converting text data into numerical format
• Handling missing or inconsistent values
• Ensuring correct data types for each column

These steps ensure that the dataset is clean and suitable for analysis and visualization.

### 4. Model Training using Azure AutoML

Automated Machine Learning (AutoML) is used to train models on the dataset. The task type selected is classification, where the model predicts transportation type based on various input features.

Different models are evaluated automatically by Azure ML, and the best-performing model is selected. In this project, the best model identified is the Voting Ensemble model, which combines multiple algorithms to improve prediction accuracy.

### 5. Data Visualization using Power BI

The cleaned dataset is imported into Power BI for visualization. Various charts and dashboards are created to represent travel data effectively.

The visualizations include:

• Bar charts to compare accommodation and transportation costs across destinations
• Pie charts to show distribution of accommodation types
• Line charts to analyze travel trends over time
• Map visualization to display geographical distribution of destinations

• Card visuals to show key metrics such as total trips and average duration

Power BI enables interactive dashboards, allowing users to filter and explore data easily.

**6. Dashboard Development**

A comprehensive dashboard is designed to present all visual insights in a single view. The dashboard includes cost analysis, duration analysis, and travel trends, making it easy to understand patterns and relationships in the data.

The final dashboard provides a clear and interactive representation of travel data, helping users make data-driven decisions.

Overall, the implementation demonstrates how cloud-based analytics and visualization tools can be integrated to analyze large datasets efficiently and generate meaningful insights.

# Experimentation and Code

The Travel Data Analysis and Recommendation System was implemented using Microsoft Azure Machine Learning and Power BI. The experimentation process involved loading the dataset, preprocessing the data, training the model, and creating visualizations to analyze travel patterns.

Code Structure

The project does not involve extensive programming as most operations are performed using cloud-based tools. However, the implementation follows a structured workflow:

• **Data Loading:** The travel dataset in CSV format is uploaded to Azure Blob Storage and accessed in Azure Machine Learning.

• **Data Preprocessing:** Data cleaning operations are performed to remove special characters such as currency symbols, handle missing values, and convert columns into appropriate data types.

• **Model Training**: Automated Machine Learning (AutoML) is used to train multiple models and evaluate their performance. The system automatically selects the best model based on accuracy.

• **Visualization**: The dataset is imported into Power BI, where different charts and dashboards are created to represent insights.

Experimentation Steps

The following steps were carried out during experimentation:

1. Uploaded the travel dataset into Azure Blob Storage.
2. Created a data asset in Azure Machine Learning.
3. Performed data preprocessing and ensured proper column formatting.
4. Configured and executed an Automated ML experiment.
5. Selected classification as the task type and chose the target column.

6. Limited the number of trials to optimize training time.
7. Trained the model and obtained the best-performing model (Voting Ensemble).
8. Imported the dataset into Power BI for visualization.
9. Cleaned the dataset using Power Query Editor in Power BI.
10. Created multiple visualizations such as bar charts, pie charts, line charts, and maps.
11. Applied filters and slicers to make the dashboard interactive.

**Performance Analysis**

• **Processing Speed**: The use of Azure ML enabled faster model training and efficient handling of the dataset.

• **Scalability**: The cloud-based system allows easy scaling for larger datasets.

• **Accuracy**: The Voting Ensemble model provided good predictive performance.

• **Resource Usage**: Minimal local resources were used since most operations were performed in the cloud.

Conclusion of Experimentation

The experimentation successfully demonstrated that cloud-based tools can efficiently process and analyze travel data. The integration of Azure Machine Learning and Power BI enabled both analytical processing and effective visualization. The system produced meaningful insights and validated the effectiveness of the approach.

## RESULTS:

The results of the Travel Data Analysis and Recommendation System demonstrate the effectiveness of cloud-based analytics and data visualization tools.

The model training process was carried out using Azure Machine Learning, where multiple models were evaluated using Automated ML. The best-performing model was identified as the Voting Ensemble model, which provided good predictive performance.

| Algorithm name | Responsible AI | AUC weighted | Sampling | Created on |
|---|---|---|---|---|
| VotingEnsemble | | 0.75167 | 100.00 % | Apr 21, 2026 9:36 PM |
| MaxAbsScaler, XGBoostClassifier | | 0.71578 | 100.00 % | Apr 21, 2026 9:33 PM |
| MaxAbsScaler, LightGBM | | 0.71345 | 100.00 % | Apr 21, 2026 9:33 PM |

Page 1 of 1 25/Page

Figure 1: Azure ML Model Result

| | | | | |
|---|---|---|---|---|
| MaxAbsScaler, ExtremeRandomTrees | | 0.70430 | 100.00 % | Apr 21, 2026 9:33 PM |
| MaxAbsScaler, RandomForest | | 0.50841 | 100.00 % | Apr 21, 2026 9:33 PM |

Page 1 of 1 25/Page

Figure 2: Azure ML Model Result

The dataset was further analyzed using Power BI, where multiple visualizations were created to extract meaningful insights.

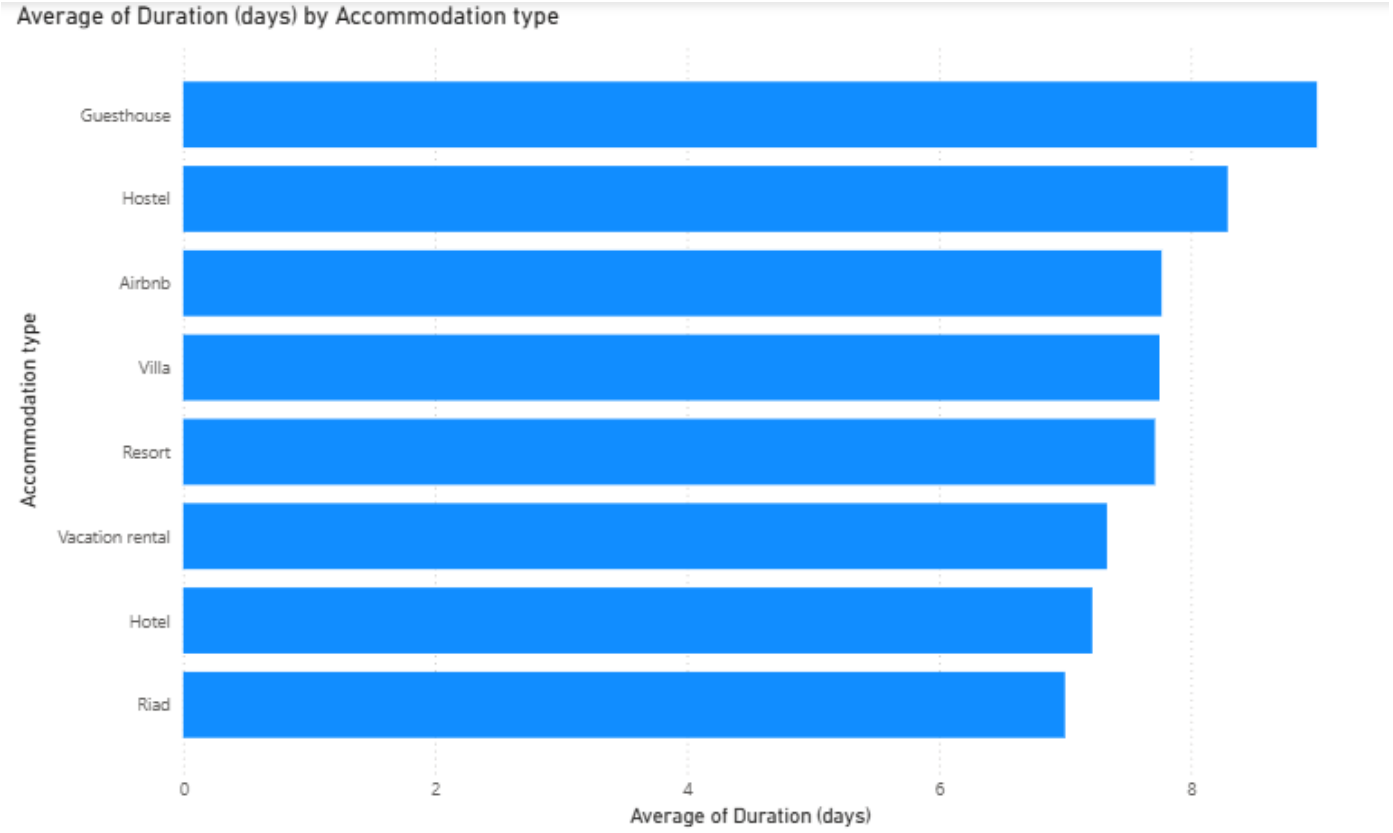



Figure 3: Accomodation Type

The analysis revealed that certain destinations have higher accommodation and transportation costs, indicating expensive travel locations.

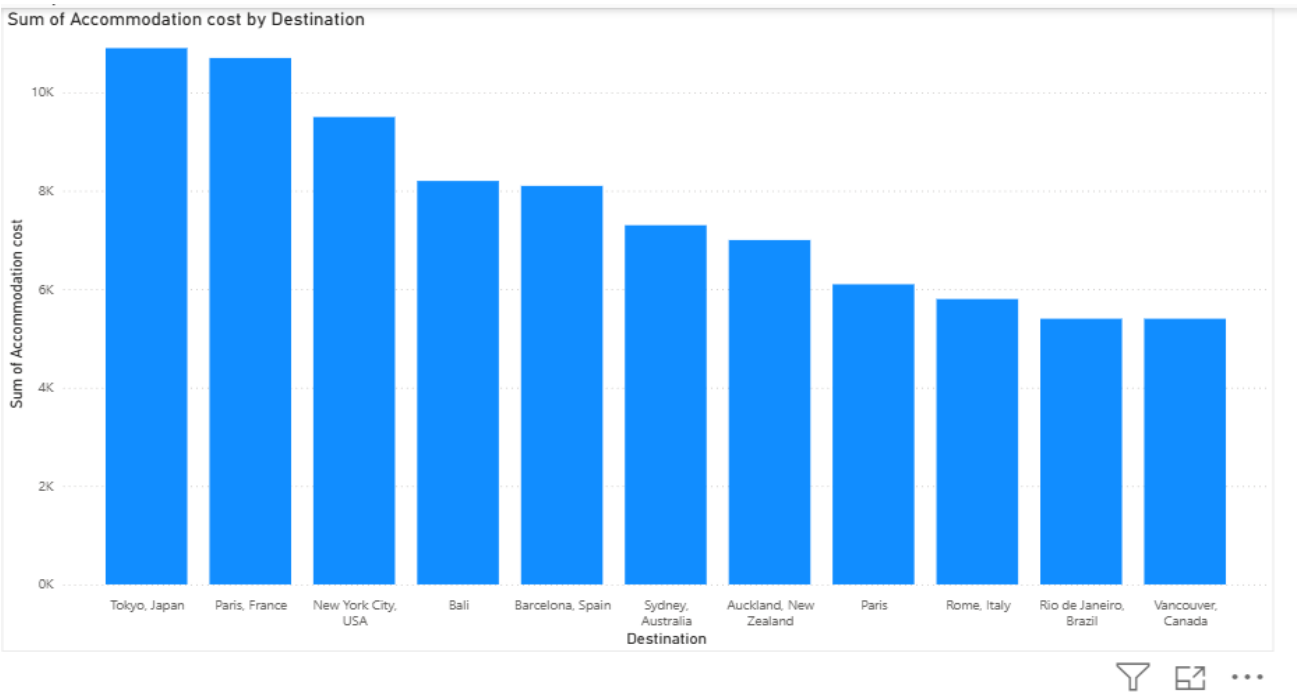
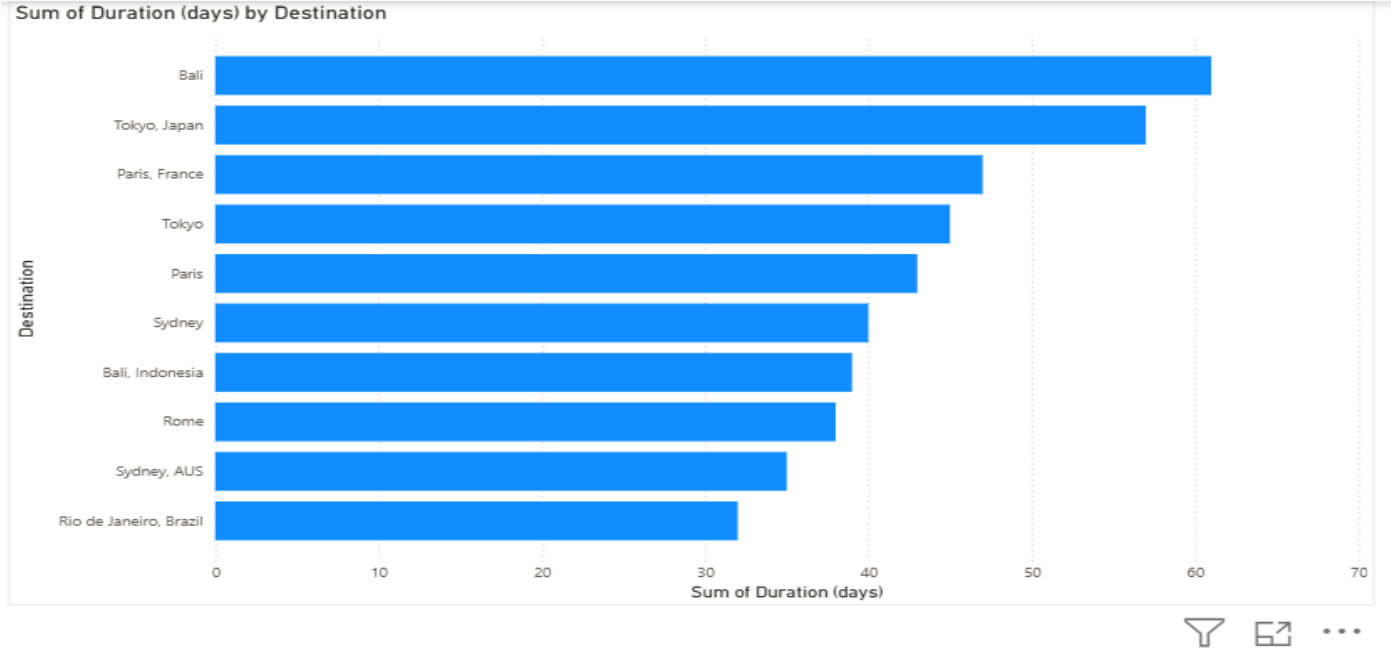
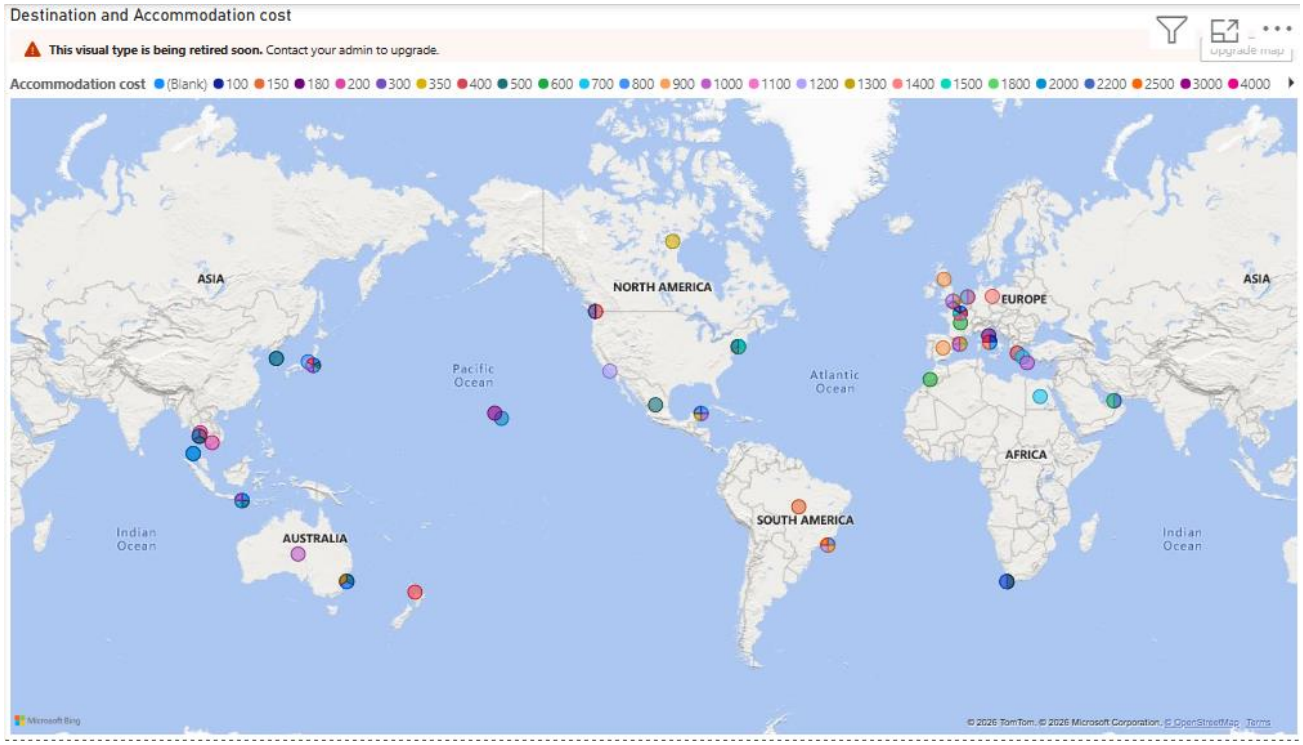



Figure 4: Accomodation Cost

The duration analysis showed that travelers tend to stay longer in certain accommodation types such as guesthouses and hostels.



The map visualization provided a geographical representation of travel destinations, showing global distribution of travel activity.




Additionally, the monthly trend analysis showed that travel activity is higher during mid-year months, indicating peak travel seasons.

Overall, the results demonstrate that the system successfully identifies travel patterns, cost distribution, and trends using cloud-based tools and visualization techniques.

# CONCLUSION

The Travel Data Analysis and Recommendation System successfully demonstrates the use of cloud-based technologies and data visualization tools to analyze travel data efficiently. By utilizing Microsoft Azure Machine Learning and Power BI, the project provides a scalable and effective solution for handling and interpreting large datasets.

The system was able to extract meaningful insights related to travel trends, cost distribution, and duration patterns. The use of Automated Machine Learning helped in identifying the best-performing model, while Power BI enabled the creation of interactive dashboards for better understanding of the data.

The visualizations clearly highlighted key observations such as peak travel months, most expensive destinations, and variations in accommodation and transportation costs. The geographical map provided a global view of travel activity, making the analysis more comprehensive and easy to interpret.

Overall, the project achieved its objectives by combining cloud computing, machine learning, and data visualization techniques. It demonstrates how data-driven approaches can support better decision-making in the travel and tourism domain.

This project can be further enhanced by incorporating real-time data, advanced machine learning models, and personalized recommendation systems to improve user experience and predictive capabilities.

# REFERENCE:

References

1. Microsoft Azure Documentation

Microsoft Azure. Azure Machine Learning Documentation.

Available at: https://learn.microsoft.com/en-us/azure/machine-learning/

This reference was used to understand model training, dataset management, and AutoML features.

2. Microsoft Power BI Documentation

Microsoft Power BI. Power BI Documentation.

Available at: https://learn.microsoft.com/en-us/power-bi/

This source was used for creating dashboards and visualizations.

3. Azure Blob Storage Documentation

Microsoft Azure. Azure Blob Storage Documentation.

Available at: https://learn.microsoft.com/en-us/azure/storage/blobs/

This reference helped in understanding cloud storage and data handling.

4. Kaggle Dataset

Kaggle. Travel Dataset.

Available at: https://www.kaggle.com/

Used as the source for travel-related data.

5. Data Visualization Concepts

Few, S. (2009). Now You See It: Simple Visualization Techniques for Quantitative Analysis.

This reference helped in understanding effective visualization techniques.